
Cloud Service Models

Cloud computing is a model that enables the end users to access the shared pool of resources such as **compute, network, storage, database**, and application as an on-demand service without the need to buy or own it.

The services are provided and managed by the **service provider**, reducing the management effort from the end user side.

The essential characteristics of the cloud include **on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service**.

The *National Institute of Standards and Technology (NIST)* defines three basic service models, namely, IaaS, PaaS, and SaaS, as shown in Figure 5.1.

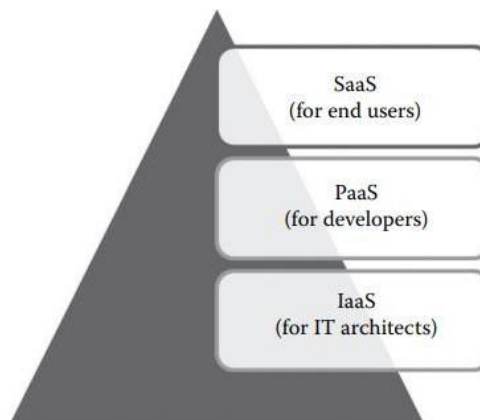


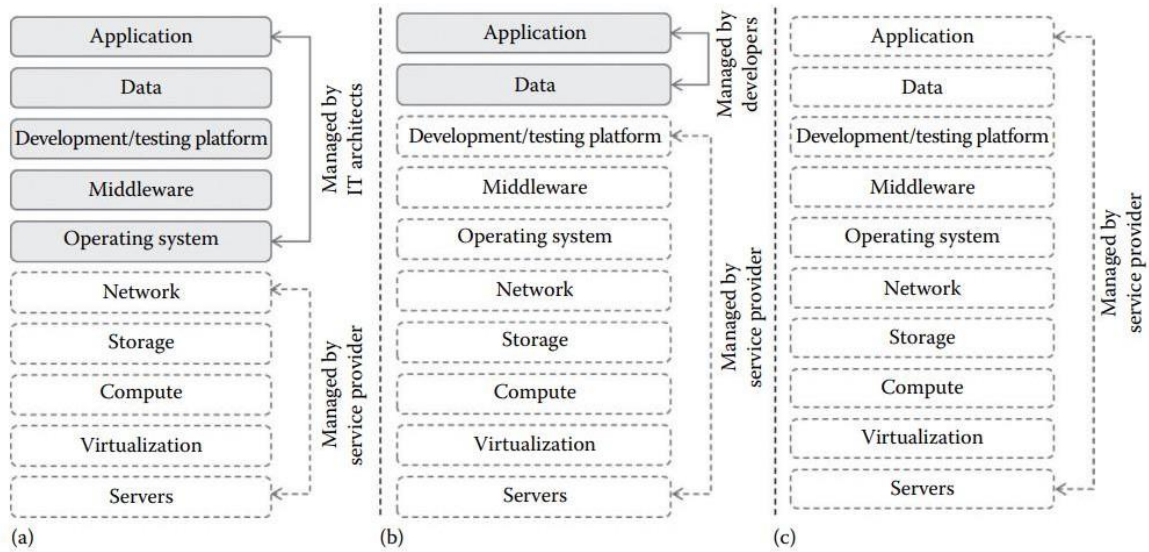
FIGURE 5.1
Basic cloud service models.

The different cloud service models target different audiences.

For example, the

- IaaS model targets the information technology (IT) architects,
- PaaS targets the developers, and
- SaaS targets the end users.

Based on the services subscribed, the responsibility of the targeted audience may vary as shown in Figure 5.2.

**FIGURE 5.2**

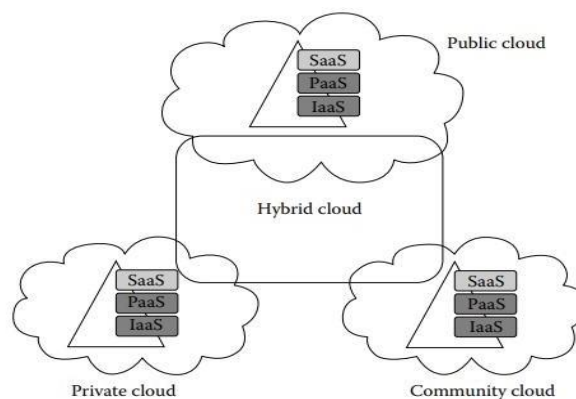
User and service provider responsibilities of cloud service models: (a) IaaS, (b) PaaS, and (c) SaaS.

The different service models of cloud computing can be deployed and delivered through any one of the cloud deployment models.

The NIST defines four different types of cloud deployment models, namely, public cloud, private cloud, community cloud, and hybrid cloud.

- The **public cloud** is provided for the general public.
- The **private cloud** is used by an organization for its multiple business units.
- The **community cloud** is for some group of organization with the same goals.
- The **hybrid cloud** is any combination of the public, private, and community clouds.

The service delivery of cloud services through different deployment models is shown in Figure 5.3.

**FIGURE 5.3**

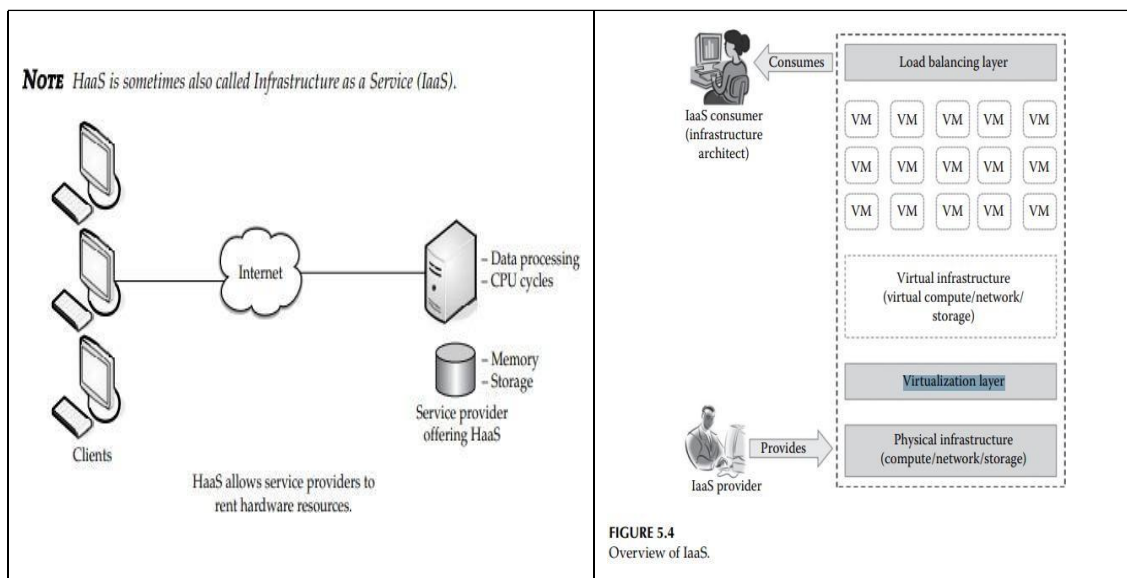
Deployment and delivery of different cloud service delivery models.

There are the following three types of cloud service models:

1. Infrastructure as a Service (IAAS)
2. Platform as a Service (PAAS)
3. Software as a Service (SAAS)

1. Infrastructure as a Service (IAAS) :

- IaaS is also known as **Hardware as a Service (HaaS)**. It is a computing infrastructure managed over the internet.
- IAAS provides businesses with virtualized computing resources over the internet.
- Infrastructure-as-a-Service (IaaS) is a cloud computing service offering **on-demand** computing, **storage**, and **networking**, **Operating Systems**, **Security** resources.
- It usually works on a **pay-as-you-go** basis.



IaaS changes the computing from a physical infrastructure to a virtual infrastructure. IaaS provides **virtual computing**, **storage**, and **network** resources.

Technology virtualization is used to provide the virtual resources. All the virtual resources are given to the virtual machines (VMs) that are configured by the service provider.

The end users or IT architects will use the infrastructure resources in the form of VMs as shown in Figure 5.4.

A typical IaaS provider may provide the following services as shown in Figure 5.5:

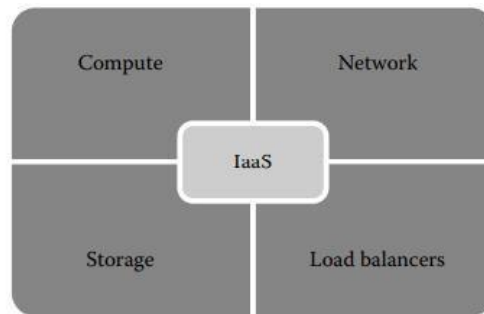


FIGURE 5.5
Services provided by IaaS providers.

1. **Compute:** Computing as a Service includes virtual central processing units (CPUs) and virtual main memory for the VMs that are provisioned to the end users.
2. **Storage:** STaaS provides back-end storage for the VM images. Some of the IaaS providers also provide the back end for storing files.
3. **Network:** Network as a Service (NaaS) provides virtual networking components such as virtual router, switch, and bridge for the VMs.
4. **Load balancers:** Load Balancing as a Service may provide load balancing capability at the infrastructure layer.

Advantages of IaaS :

There are the following advantages of IaaS computing layer -

1. Shared infrastructure

IaaS allows multiple users to share the same physical infrastructure.

2. Web access to the resources

IaaS allows IT users to access resources over the internet.

3. Pay-as-per-use model

IaaS providers provide services based on the pay-as-per-use basis. The users are required to pay for what they have used.

4. On-demand scalability

On-demand scalability is one of the biggest advantages of IaaS. Using IaaS, users do not worry about to upgrade software and troubleshoot the issues related to hardware components.

Disadvantages of IaaS :**1. Security**

Security is one of the biggest issues in IaaS. Most of the IaaS providers are not able to provide 100% security.

2. Maintenance & Upgrade

Although IaaS service providers maintain the software, but they do not upgrade the software for some organizations.

3. Interoperability issues OR vendor lock-in

It is difficult to migrate VM from one IaaS provider to the other, so the customers might face problem related to vendor lock-in.

TABLE 5.1

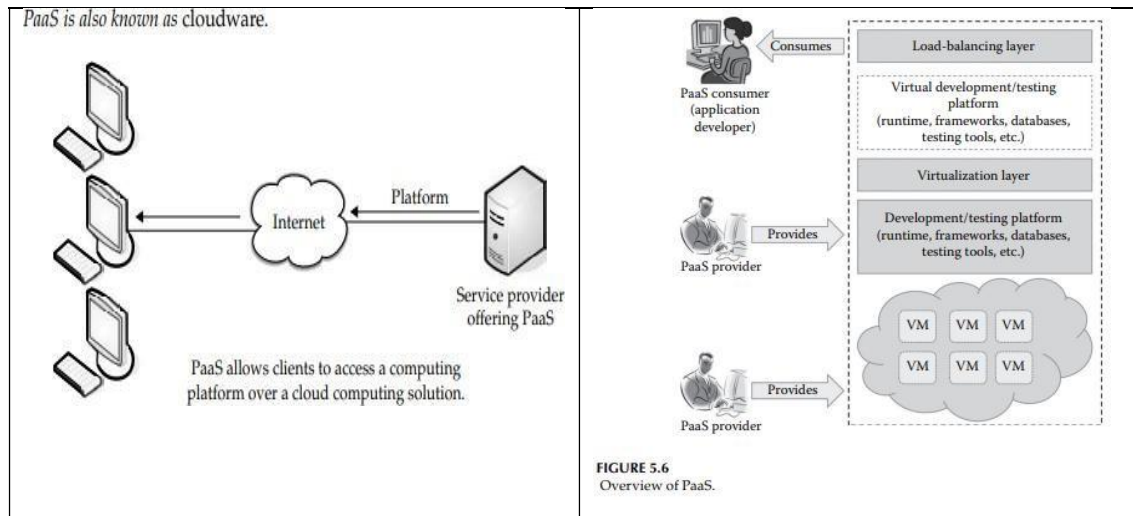
Summary of Popular IaaS Providers

Provider	License	Deployment Model	Host OS	Guest OS	Supported Hypervisor(s)
Amazon Web Services	Proprietary	Public	Not available	Red Hat Linux, Windows Server, SuSE Linux, Ubuntu, Fedora, Debian, CentOS, Gentoo Linux, Oracle Linux, and FreeBSD	Xen
Google Compute Engine	Proprietary	Public	Not available	Debian 7 Wheezy, CentOS 6, Red Hat Enterprise Linux, SUSE, Windows Server, CoreOS, FreeBSD, and SELinux	KVM
Microsoft Windows Azure	Proprietary	Public	Not available	Windows Server, CentOS, FreeBSD, openSUSE Linux, and Oracle Enterprise Linux	Windows Azure hypervisor
Eucalyptus	GPLv3	Private and hybrid	Linux	Linux and Windows	Xen, KVM, VMware
Apache CloudStack	Apache 2	Private	Linux	Windows, Linux, and various versions of BSD	KVM, vSphere, XenServer/XCP
OpenNebula	Apache 2	Private, public, and hybrid	CentOS, Debian, and openSUSE	Microsoft Windows and Linux	Xen, KVM, VMware
OpenStack	Apache 2	Private and public	CentOS, Debian, Fedora, RHEL, openSUSE, and Ubuntu	CentOS, Ubuntu, Microsoft Windows, and FreeBSD	libvirt, Hyper-V, VMware, XenServer 6.2, baremetal, docker, Xen, LXC via libvirt

2. Platform as a Service (PAAS) :

- **PaaS** supplies all the resources required to build applications and services completely from the Internet, without having to download or install software.
- **Platform as a Service (PaaS)** is a way to build applications and have them hosted by the cloud provider.
- It allows you to deploy applications without having to spend the money to buy the servers on which to house them.
- PaaS allows the developers to develop their application online and also allows them to deploy immediately on the same platform.
- PaaS consumers or developers can consume *language runtimes, application frameworks, databases, message queues, testing tools, and deployment tools as a service over the Internet.*

- Thus, it reduces the complexity of buying and maintaining different tools for developing an application.



- Typical PaaS providers may provide programming languages, application frameworks, databases, and testing tools as shown in Figure 5.7.
- Some of the PaaS providers also provide **build tools, deployment tools, and software load balancers as a service**:

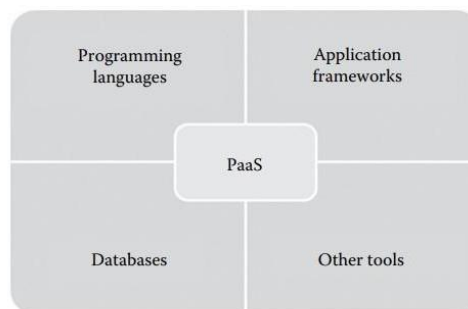


FIGURE 5.7
Services provided by PaaS providers.

- 1. Programming languages:** PaaS providers provide a wide variety of programming languages for the developers to develop applications. Some of the popular programming languages provided by PaaS vendors are *Java, Perl, PHP, Python, Ruby, Scala, Clojure, and Go*.
- 2. Application frameworks:** PaaS vendors provide application frameworks that simplify the application development. Some of the popular application development frameworks provided by a PaaS provider include *Node.js, Rails, Drupal, Joomla, WordPress, Django, EE6, Spring, Play, Sinatra, Rack, and Zend*.
- 3. Database:** Since every application needs to communicate with the databases, it becomes a must-have tool for every application.

PaaS providers are providing databases also with their PaaS platforms. The popular databases provided by the popular PaaS vendors are ***ClearDB, PostgreSQL, Cloudant, Membase, MongoDB, and Redis.***

4. **Other tools:** PaaS providers provide all the tools that are required to develop, test, and deploy an application.

Advantages of PaaS:

1. **All in one:** Most of the PaaS providers offer services to develop, test, deploy, host, and maintain applications in the same IDE.
2. **Simple and convenient for users:** It provides much of the infrastructure and other IT services, which users can access anywhere via a web browser.
3. **Cost-Effective:** It charges for the services provided on a per-use basis thus eliminating the expenses one may have for on-premises hardware and software.
4. **Efficiently managing the lifecycle:** It is designed to support the complete web application lifecycle: building, testing, deploying, managing, and updating.
5. **Efficiency:** It allows for higher-level programming with reduced complexity thus, the overall development of the application can be more effective.

Disadvantages of Paas:

1. **Limited control over infrastructure:** PaaS providers typically manage the underlying infrastructure and take care of maintenance and updates, but this can also mean that users have less control over the environment and may not be able to make certain customizations.
2. **Dependence on the provider:** Users are dependent on the PaaS provider for the availability, scalability, and reliability of the platform, which can be a risk if the provider experiences outages or other issues.
3. **Limited flexibility:** PaaS solutions may not be able to accommodate certain types of workloads or applications, which can limit the value of the solution for certain organizations.

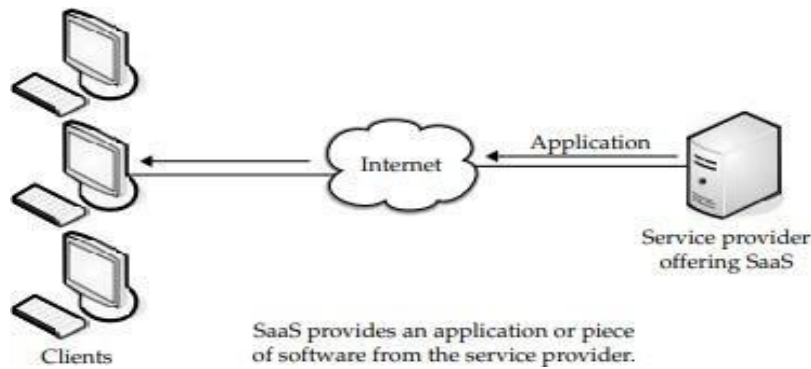
TABLE 5.2

Summary of Popular PaaS Providers

Provider	License	Deployment Model	Supported Languages	Supported Frameworks	Supported Databases	Client Tools
Cloud Foundry	Open source and proprietary	Public	Python, PHP, Java, Groovy, Scala, and Ruby	Spring, Grails, Play, Node.js, Lift, Rails, Sinatra, and Rack	MySQL, PostgreSQL, MongoDB, and Redis	cf. CLI, IDEs, and build tools
Google App Engine	Proprietary	Public	Python, Java, Groovy, JRuby, Scala, Clojure, Go, and PHP	Django, CherryPy, Pyramid, Flask, web2py, and webapp2.	Google Cloud SQL, Datastore, BigTable, and Blobstore	APIs
Heroku	Proprietary	Public	Ruby, Java, Scala, Clojure and Python, PHP, and Perl	Rails, Play, Django, and Node.js.	ClearDB, PostgreSQL, Cloudant, Membase, MongoDB, and Redis	CLI and RESTful API
Microsoft Windows Azure	Proprietary	Public	.Net, PHP, Python, Ruby, and Java	Django, Rails, Drupal, Joomla, WordPress, DotNetNuke, and Node.js.	SQL Azure, MySQL, MongoDB, and CouchDB	RESTful API and IDEs

3. Software as a Service (SAAS) :

- **Software as a Service (SaaS)** is the model in which an application is hosted as a service to customers who access it via the Internet.
- SaaS is simply the cloud vendor providing the given piece of software you want to use, on their server.
- Software-as-a-Service (SaaS) is a way of delivering services and applications over the Internet. Instead of installing and maintaining software, we simply access it via the Internet, freeing ourselves from the complex software and hardware management.
- It removes the need to install and run applications on our own computers or in the data centers eliminating the expenses of hardware as well as software maintenance.
- SaaS provides a complete software solution that you purchase on a **pay-as-you-go** basis from a cloud service provider.
- Most SaaS applications can be run directly from a web browser without any downloads or installations required.
- The SaaS applications are sometimes called **Web-based software, on-demand software, or hosted software**.



A typical SaaS provider may provide business services, social networks, document management, and mail services as shown in Figure 5.8:

1. **Business services:** Most of the SaaS providers started providing a variety of business services that attract start-up companies. The business SaaS services include ERP, CRM, billing, sales, and human resources.
2. **Social networks:** Since social networking sites are extensively used by the general public, many social networking service providers adopted SaaS for their sustainability. Since the number of users of the social networking sites is increasing exponentially, cloud computing is the perfect match for handling the variable load.

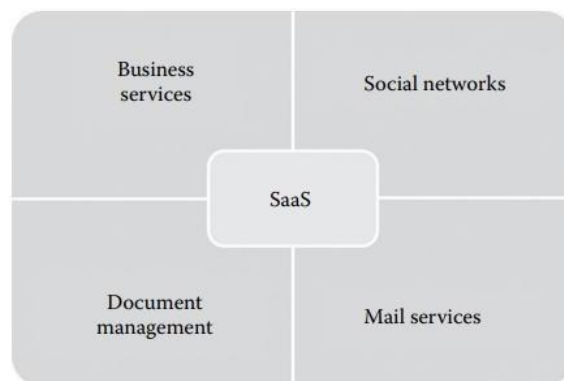


FIGURE 5.8
Services provided by SaaS Providers.

3. **Document management:** Since most of the enterprises extensively use electronic documents, most of the SaaS providers started providing services that are used to create, manage, and track electronic documents.
4. **Mail services:** E-mail services are currently used by many people. The future growth in e-mail usage is unpredictable. To handle the unpredictable number

of users and the load on e-mail services, most of the e-mail providers started offering their services as SaaS services.

There are many types of software in SaaS model.

- Customer resource management (CRM)
- Video conferencing
- IT service management
- Accounting
- Web analytics
- Web content management

Advantages of SaaS

- **No client-side installation:** SaaS services do not require client-side installation of the software. The end users can access the services directly from the service provider data center without any installation.
- **Cost-Effective:** Pay only for what you use.
- **One to many:** SaaS services are delivered as a one-to-many model where a single instance of the application can be shared by multiple tenants or customers
- **Reduced time:** Users can run most SaaS apps directly from their web browser without needing to download and install any software. This reduces the time spent in installation and configuration and can reduce the issues that can get in the way of the software deployment.
- **Accessibility:** We can Access app data from anywhere.
- **Automatic updates:** Rather than purchasing new software, customers rely on a SaaS provider to automatically perform the updates.
- **Scalability:** It allows the users to access the services and features on-demand.

Disadvantages of Saas:

- **Limited customization:** SaaS solutions are typically not as customizable as on-premises software, meaning that users may have to work within the constraints of the SaaS provider's platform.
- **Dependence on internet connectivity:** SaaS solutions are typically cloud-based, which means that they require a stable internet connection to function properly. This can be problematic for users in areas with poor connectivity or for those who need to access the software in offline environments.

- **Security concerns:** SaaS providers are responsible for maintaining the security of the data stored on their servers, but there is still a risk of data breaches or other security incidents.
- **Limited control over data:** SaaS providers may have access to a user's data, which can be a concern for organizations that need to maintain strict control over their data for regulatory or other reasons.

TABLE 5.3

Summary of Popular SaaS Providers

Provider	Services Provided
Salseforce.com	On-demand CRM solutions
Google Apps	Gmail, Google Calendar, Talk, Docs, and Sites
Microsoft Office 356	Online office suite, software, plus services
NetSuite	ERP, accounting, order management, inventory, CRM, professional services automation (PSA), and e-commerce applications
Concur	Integrated travel and expense management solutions
GoToMeeting	Online meeting, desktop sharing, and video-conferencing software
Constant Contact	E-mail marketing, social-media marketing, online survey, event marketing, digital storefronts, and local deals tools
Workday, Inc.	Human capital management, payroll, and financial management
Oracle CRM	CRM applications
Intacct	Financial management and accounting software solutions

Comparison of different Service Models

Difference between IaaS, PaaS, and SaaS

IaaS	Paas	SaaS
It provides a virtual data center to store information and create platforms for app development, testing, and deployment.	It provides virtual platforms and tools to create, test, and deploy apps.	It provides web software and apps to complete business tasks.
It provides access to resources such as virtual machines, virtual storage, etc.	It provides runtime environments and deployment tools for applications.	It provides software as a service to the end-users.
It is used by network architects.	It is used by developers.	It is used by end users.
IaaS provides only Infrastructure.	PaaS provides Infrastructure+Platform.	SaaS provides Infrastructure+Platform +Software.

OR

Aspect	Infrastructure as a Service (IaaS)	Platform as a Service (PaaS)	Software as a Service (SaaS)
Definition	Provides virtualized computing resources over the internet. Includes virtual machines, storage, and networking. Users manage the OS and applications.	Offers a platform that includes not only infrastructure but also development tools and services. Users focus on application development while the platform handles	Delivers software applications over the internet on a subscription basis. Users access applications through a web browser.
		infrastructure.	
Management	Users have more control over the infrastructure, including the OS, applications, and security.	Users focus on application development and deployment, leaving infrastructure management to the provider.	Users have minimal control over the underlying infrastructure, focusing solely on using the application.
Scalability	Offers scalability for infrastructure resources, allowing users to scale up or down based on their requirements.	Provides scalability for applications and services, often within the constraints of the underlying platform.	Scalability is typically managed by the SaaS provider, and users can scale the number of users or resources based on their subscription.
Development	Suited for organizations with in-house IT expertise and the need for full infrastructure control.	Ideal for developers who want to focus on application development without worrying about infrastructure.	Targeted at end-users and businesses looking for ready-to-use applications.
Examples	Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform.	Heroku, Google App Engine, Microsoft Azure App Service.	Salesforce, Dropbox, Microsoft Office 365.
Use Cases	Good for running custom applications, hosting websites, and managing virtual servers.	Suitable for web and mobile app development, where the focus is on coding, not infrastructure.	Used for email, collaboration tools, CRM, and a wide range of software applications.
Cost	Costs may vary based on the resources used. Users pay for infrastructure components (VMs, storage, etc.).	Typically, a subscription or consumption-based model, often more cost-effective as infrastructure management is abstracted.	Usually subscription-based, with a fixed monthly or annual fee.
Security and Compliance	Users have more control and responsibility for securing the infrastructure and data.	Security responsibilities are shared between the provider and the user.	Providers are responsible for securing both the application and the underlying infrastructure.
Customization	Offers high customization, allowing users to configure the	Customization is limited to the platform's capabilities.	Limited customization options, as users typically configure the

Cloud Deployment Models

What is a deployment model?

It defines the type of access to the cloud, how much data you want to store and who has access to the Infrastructure and that is the how cloud is located?

Introduction:

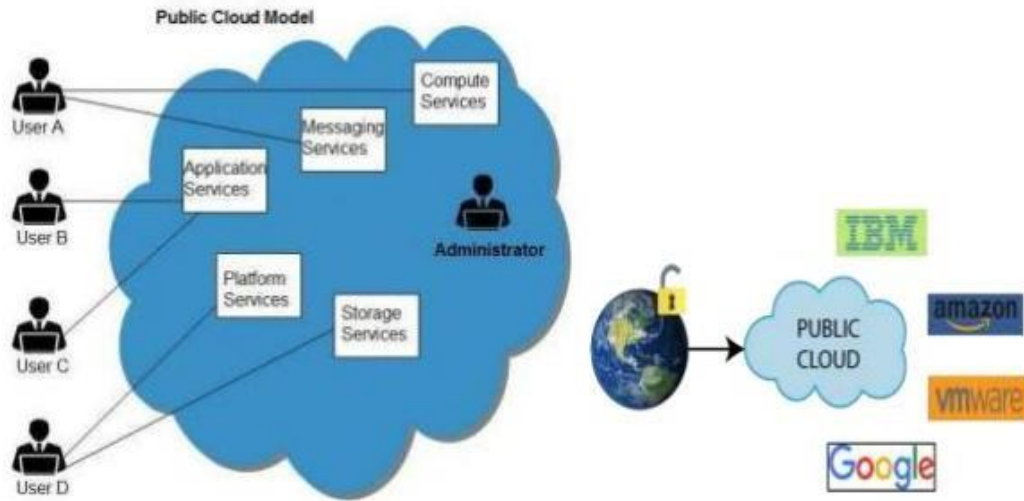
- It is important to have an idea about the deployment models because setting up a cloud is the most basic requirement prior to starting any further study about cloud computing.
- In the business perspective, making the correct decision regarding the deployment model is very important.
- A model should be selected based on the needs, requirements, budget, and security.
- A wrong decision in the deployment model may affect the organization heavily.
- Hence, it is very important to know about deployment models.
- There are many users of the cloud, and each user has different needs.
- One deployment model will not suite all the cloud users.
- Based on the cloud setup, the properties of the cloud change.
- There are four types of deployment models available in the cloud, namely, **private, public, community, and hybrid.**

1. Public cloud:

Explain Public Cloud with neat diagram

- Public Cloud allows systems and services to be easily accessible to general public.
- The IT giants such as Google, Amazon and Microsoft offer cloud services via Internet.
- The Public Cloud Model is shown in the diagram below.
- Public Cloud provides a shared platform that is accessible to the general public through an Internet connection.
- Public cloud operated on the pay-as-per-use model and administrated by the third party, i.e., Cloud service provider.
- In the Public cloud, the same storage is being used by multiple users at the same time.
- Public cloud is owned, managed, and operated by businesses, universities, government organizations, or a combination of them.

- Amazon Elastic Compute Cloud (EC2), Microsoft Azure, IBM's Blue Cloud, Sun Cloud, and Google Cloud are examples of the public cloud.



- According to **NIST**, the public cloud is the cloud infrastructure that is provisioned (condition/terms) for open use by the general public.
- It may be owned, managed, and operated by a business, academic, or government organization, or some combination of them.

Characteristics

Highly scalable:

- The public cloud is highly scalable.
- The resources in the public cloud are large in number and the service providers make sure that all the requests are granted.
- Hence, the public cloud is considered to be scalable.

Affordable:

- The public cloud is offered to the public on a pay-as-you-go basis;
- Hence, the user has to pay only for what he or she is using (usually on a per-hour basis).
- And, this does not involve any cost related to the deployment. Less secure:
- The public cloud is less secure out of all the four deployment models.
- This is because the public cloud is offered by a third party and they have full control over the cloud.
- Though the SLAs ensure privacy, still there is a high risk of data being leaked.

Highly available:

- The public cloud is highly available because anybody from any part of the world can access the public cloud with proper permission..

- And this is not possible in other models as geographical or other access restrictions might be there.

Stringent SLAs:

- SLA is very stringent in the case of the public cloud.
- As the service provider's business reputation and customer strength are totally dependent on the cloud services.
- They follow the SLA strictly and violations are avoided. These SLAs are very competitive.

Advantages

- There is no need of establishing infrastructure for setting up a cloud.
- There is no need for maintaining the cloud.
- They are comparatively less costly than other cloud models.
- Strict SLAs are followed.
- There is no limit for the number of users.
- The public cloud is highly scalable.
- **Low Cost**- Public cloud has a lower cost than private, or hybrid cloud, as it shares the same resources with a large number of consumers.
- **Location Independent**- Public cloud is location independent because its services are offered through the internet.
- **Save Time**- In Public cloud, the cloud service provider is responsible for the manage and maintain data centres in which data is stored, so the cloud user can save their time to establish connectivity, deploying new products, release product updates, configure, and assemble servers.
- **Quickly and easily set up**- Organizations can easily buy public cloud on the internet and deployed and configured it remotely through the cloud service provider within a few hours.
- **Scalability and reliability**- Public cloud offers scalable (easy to add and remove) and reliable (24*7 available) services to the users at an affordable cost.

Disadvantages

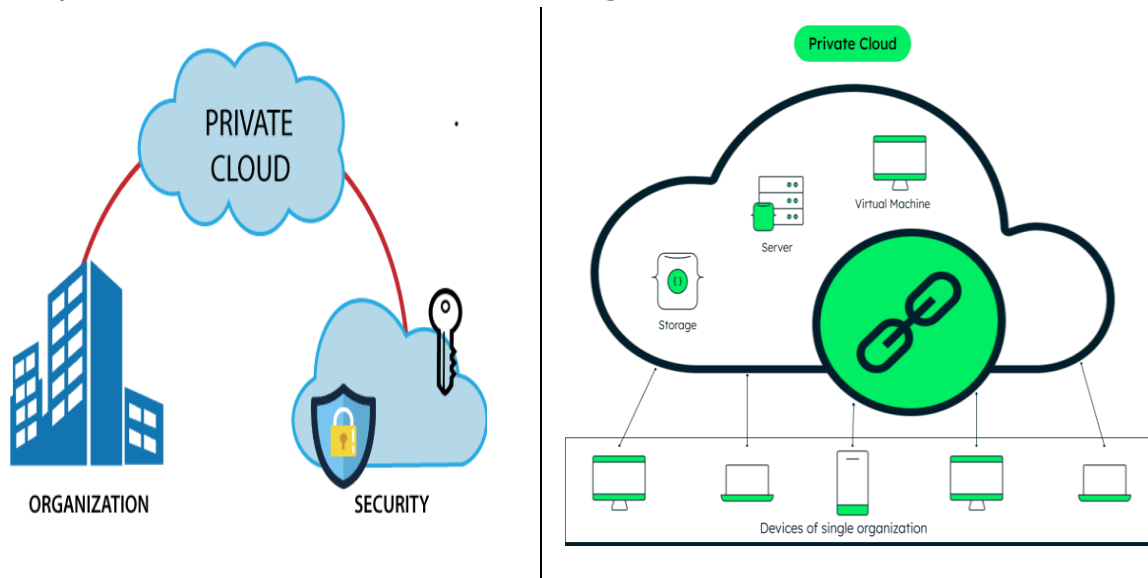
- **Security** is an issue.
- **Privacy** and organizational autonomy are not possible.
- **Low Security**- Public Cloud is less secure because resources are shared publicly.
- **Performance**- In the public cloud, performance depends upon the speed of internet connectivity.
- **Less customizable**-Public cloud is less customizable than the private cloud.

2. Private cloud

- Private cloud is also known as an internal cloud or corporate cloud.
- Private cloud provides computing services to a private internal network (within the organization) and selected users instead of the general public.
- Private cloud provides a high level of security and privacy to data through firewalls and internal hosting.

- *HP Data Centers, Microsoft, Elastra-private cloud, and Ubuntu* are the example of a private cloud.
- Private Cloud allows systems and services to be accessible within an organization. The Private Cloud is operated only within a single organization.
- However, it may be managed internally by the organization itself or by third-party.

The private cloud model is shown in the diagram below



Certain characteristics of the private cloud are as follows:

Secure:

The private cloud is secure. This is because usually the private cloud is deployed and managed by the organization itself, and hence there is least chance of data being leaked out of the cloud.

Central control:

The organization mostly has full control over the cloud as usually the private cloud is managed by the organization itself. Thus, when managed by the organization itself, there is no need for the organization to rely on anybody.

Weak SLAs:

Formal SLAs may or may not exist in a private cloud. But if they exist they are weak as it is between the organization and the users of the same organization. Thus, high availability and good service may or may not be available. This depends on the organization that is controlling the cloud.

According to NIST(National Institute of Standards and Technology), the private cloud can be classified into several types based on their location and management:

- **On-premise private cloud**
- **Outsourced private cloud**

On-Premise Private Cloud

- On-premise private cloud is a typical private cloud that is managed by a single organization.
- Here, the cloud is deployed in organizational premises and is connected to the organizational network.

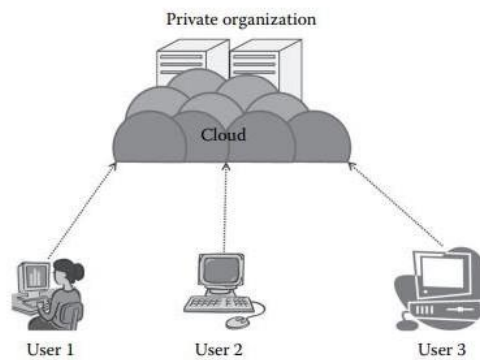


FIGURE 4.2
On-premise private cloud.

Outsourced Private Cloud

- The outsourced private cloud has a cloud outsourced to a third party
- A third party manages the whole cloud.
- Everything is same as usual private cloud except that here the cloud is outsourced.

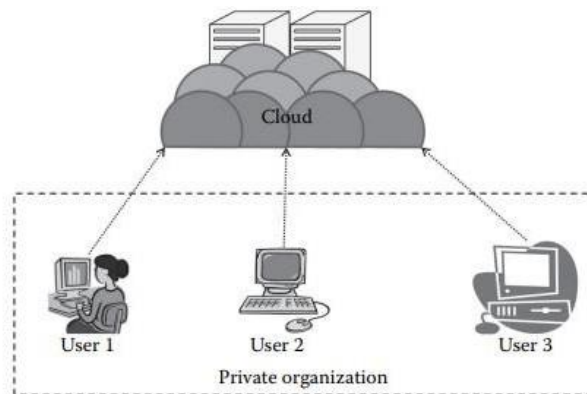


FIGURE 4.3
Outsourced private cloud.

There are several advantages and disadvantages of a private cloud.

Advantages

- The cloud is small in size and is easy to maintain.
- It provides a high level of security and privacy to the user.
- It is controlled by the organization.
- **More Control-** Private Clouds have more control over their resources and hardware than public clouds because it is only accessed by selected users.

- **Security & privacy**- Security & privacy are one of the big advantages of cloud computing. Private cloud improved the security level as compared to the public cloud.
- **Improved performance**- Private cloud offers better performance with improved speed and space capacity.

Disadvantages

- For the private cloud, budget is a constraint.
- The private clouds have loose **SLAs**.
- **High cost**- The cost is higher than a public cloud because set up and maintain hardware resources are costly.
- **Restricted area of operations**- As we know, private cloud is accessible within the organization, so the area of operations is limited.
- **Limited scalability**- Private clouds are scaled only within the capacity of internal hosted resources.
- **Skilled people**- Skilled people are required to manage and operate cloud services.

3. Hybrid Cloud

- Community Cloud allows system and services to be accessible by group of organizations.
- It shares the infrastructure between several organizations from a specific community.
- It may be managed internally by organizations or by the third-party.
- The Community Cloud Model is shown in the diagram below.

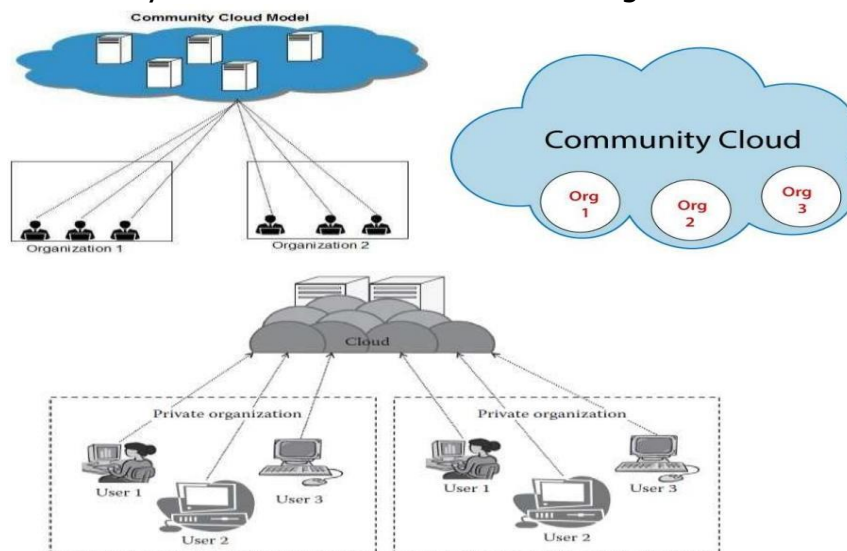


FIGURE 4.5
Community cloud.

- The hybrid cloud can be regarded as a private cloud extended to the public cloud. This aims at utilizing the power of the public cloud by retaining the properties of the private cloud.

- One of the popular examples for the hybrid cloud is ***Eucalyptus***.
- ***Eucalyptus*** was initially designed for the private cloud and is basically a private cloud, but now it also supports hybrid cloud.
- Above Figure shows the hybrid cloud. Eucalyptus is paid and open-source computer software for building Amazon Web Services-compatible private and hybrid cloud computing environments, originally developed by the company Eucalyptus Systems

Characteristics**Scalable:**

- The hybrid cloud is a combination of one or more deployment models.
- Usually, the private with public cloud gives hybrid cloud.
- The main reason of having a hybrid cloud is to use the property of a public cloud with a private cloud environment.
- The public cloud is used whenever needed; hence, as the public cloud is scalable, the hybrid cloud with the help of its public counterpart is also scalable.

Partially secure:

- The hybrid cloud usually is a combination of public and private.
- The private cloud is considered to be secured, but as the hybrid cloud also uses the public cloud, there is high risk of security breach. Thus, it cannot be fully termed as secure but as partially secure.

Stringent SLAs:

- As the hybrid cloud involved a public cloud intervention.
- the SLAs are stringent and might as per the public cloud service provider.

Complex cloud management:

- Cloud management is complex and is a difficult task in the hybrid cloud as it involves more than one type of deployment models and also the numbers of users are high.

Advantages

- It gives the power of both the private and public clouds.
- It is highly scalable. • It provides better security than the public cloud.
- Flexible and secure
- It provides flexible resources because of the public cloud and secure resources because of • the private cloud.
- Cost effective: Hybrid cloud costs less than the private cloud. It helps organizations to save costs for • both infrastructure and application support.
- It offers the features of both the public as well as the private cloud.

- A hybrid cloud is capable of adapting to the demands that each company needs for space, memory, and system.
- Security: Hybrid cloud is secure because critical activities are performed by the private cloud.
- Risk Management: Hybrid cloud provides an excellent way for companies to manage the risk.

Disadvantages

- The security features are not as good as the public cloud.
- Managing a hybrid cloud is complex.
- It has stringent SLAs.
- Networking issues: In the Hybrid Cloud, networking becomes complex because of the private and the public cloud.
- Reliability: The reliability of the services depends on cloud service providers.

4. Community Cloud

- Community Cloud allows system and services to be accessible by group of organizations.
- It shares the infrastructure between several organizations from a specific community.
- It may be managed internally by organizations or by the third-party.
- The Community Cloud Model is shown in the diagram below.

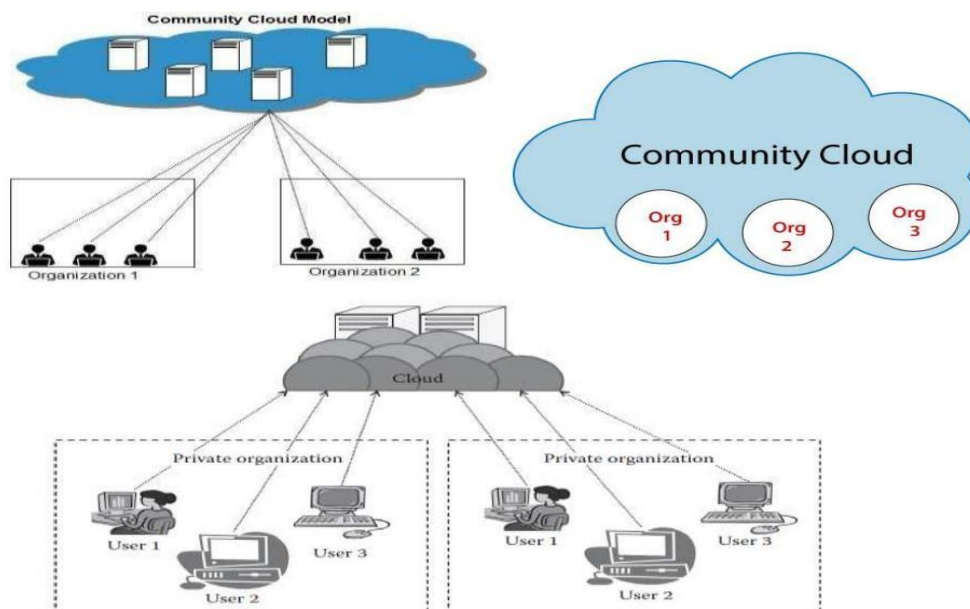


FIGURE 4.5
Community cloud.

- It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination of them, and it may exist on or off premises.
- It is a further extension of the private cloud.

- Here the organizations are able to extract the power of the cloud, which is much bigger than the private cloud, and at the same time, they are able to use it at a usually less cost.
- It is owned, managed, and operated by one or more organizations in the community, a third party, or a combination of them.

Advantages

- It allows establishing a low-cost private cloud.
- It allows collaborative work on the cloud.
- It allows sharing of responsibilities among the organization.
- It has better security than the public cloud.
- Cost effective- Community cloud is cost effective because the whole cloud is shared between several organizations or a community.
- Flexible and Scalable- The community cloud is flexible and scalable because it is compatible with every user. It allows the users to modify the documents as per their needs and requirement. Security-Community cloud is more secure than the public cloud but less secure than the private cloud.
- Sharing infrastructure-Community cloud allows us to share cloud resources, infrastructure, and other capabilities among various organizations.

Disadvantages

- Autonomy of an organization is lost.
- Security features are not as good as the private cloud.
- It is not suitable if there is no collaboration.
- Community cloud is not a good choice for every organization.
- The fixed amount of data storage and bandwidth is shared among all community members.
- Community Cloud is costly than the public cloud.
- Sharing responsibilities among organizations is difficult.
- ISSUES- Since all data is located at one place, one must be careful in storing data in community cloud because it might be accessible to others.
- It is also challenging to allocate responsibilities of governance, security and cost among organizations.

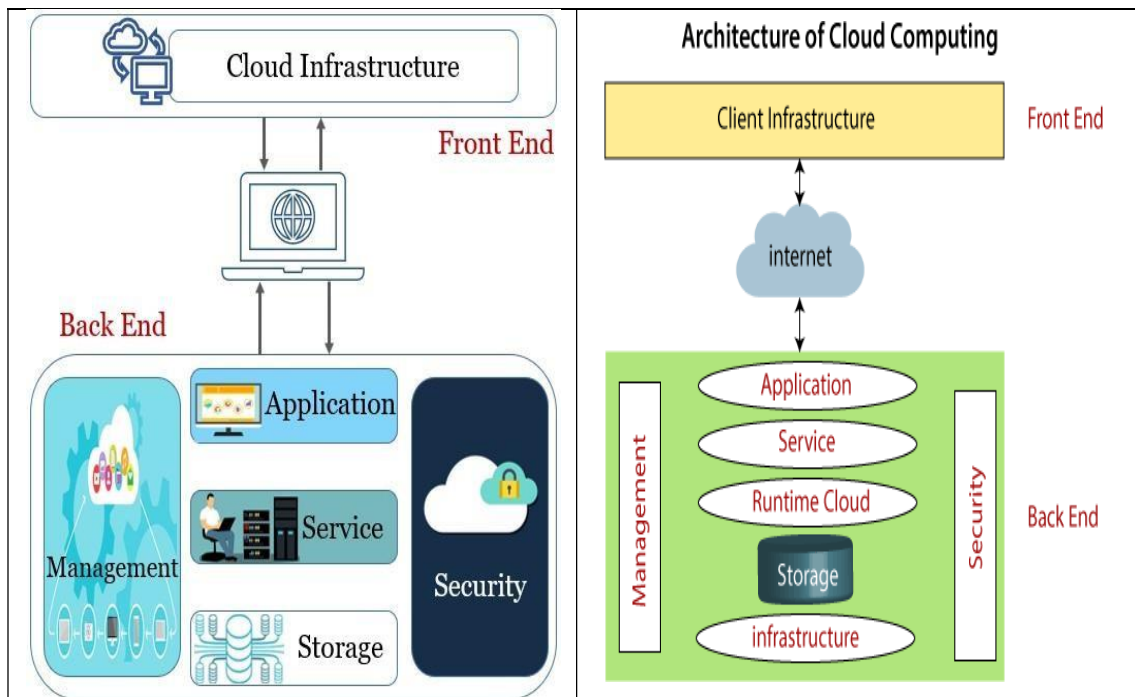
Cloud Computing Architecture- Layered Architecture of Cloud.

Cloud Computing Architecture;

- Cloud architecture defines the working mechanism of cloud computing.
- The architecture details about all the components that are essential for the working of the cloud.
- It helps us in understanding how cloud services are made available to the end users.

What is Cloud Architecture?

- In a generalized way, we can classify the cloud architecture into two ends i.e. front end and the back end.
- The front end is the one with which the end users interact.
- The back end is one with which the cloud provider has to deal.
- Both of these ends are connected to each other with the help of the internet.
- The Internet is the most crucial element of cloud computing as every user needs the internet if it has to access the cloud.
- Each of the ends is connected through a network, usually Internet.
- Cloud Architecture Diagram The figure below will show you the separation between the front and back end.



Front End

- The front end of the cloud architecture consists of applications and interfaces.
- They are required to access cloud-based services.
- The main and the only component of the front end is the cloud infrastructure.
- The front end refers to the client part of cloud computing system.
- It consists of interfaces and applications that are required to access the cloud computing platforms, Example - Web Browser.

The infrastructure consists of hardware resources that have the capability to connect to the internet such as mobile devices, tablets, and laptops.

The applications used for initiating access to the cloud are web browsers like:

- Microsoft Edge
- Google Chrome
- Firefox etc..

Back End

- The back end of the cloud architecture consists of several components as discussed below.
- The back End refers to the cloud itself.
- It consists of all the resources required to provide cloud computing services.
- It comprises of huge data storage, virtual machines, security mechanism, services, deployment models, servers, etc.

Components of Cloud Computing Architecture

There are the following components of cloud computing architecture –

1. Client Infrastructure:

- Client Infrastructure is a Front end component. It provides GUI (Graphical User Interface) to interact with the cloud.

2. Application:

- The application may be any software or platform that a client wants to access. It Provides software or the platform that the client actually wants to access. The

clients can access these applications with the help of the resources present at the back end.

3. **Service:**

A Cloud Services manages that which type of service you access according to the client's requirement.

Cloud computing offers the following three type of services:

*i. **Software as a Service (SaaS)*** – It is also known as cloud application services. Mostly, SaaS applications run directly through the web browser means we do not require to download and install these applications.

Some important example of SaaS is given below –

Example: Google Apps, Salesforce Dropbox, Slack, Hubspot, Cisco WebEx.

*ii. **Platform as a Service (PaaS)*** – It is also known as cloud platform services. It is quite similar to SaaS, but the difference is that PaaS provides a platform for software creation, but using SaaS, we can access software over the internet without the need of any platform.

Example: Windows Azure, Force.com, Magento Commerce Cloud, OpenShift.

*iii. **Infrastructure as a Service (IaaS)*** – It is also known as cloud infrastructure services. It is responsible for managing applications data, middleware, and runtime environments.

Example: Amazon Web Services (AWS) EC2, Google Compute Engine (GCE), Cisco Metapod.

4. **Runtime Cloud:**

Runtime Cloud provides the execution and runtime environment to the virtual machines.

5. **Storage Storage:**

Storage is one of the most important components of cloud computing. It provides a huge amount of storage capacity in the cloud to store and manage data.

6. **Infrastructure:**

It provides services on the host level, application level, and network level. Cloud infrastructure includes hardware and software components such as servers, storage, network devices, virtualization software, and other storage resources that are needed to support the cloud computing model.

7. Management:

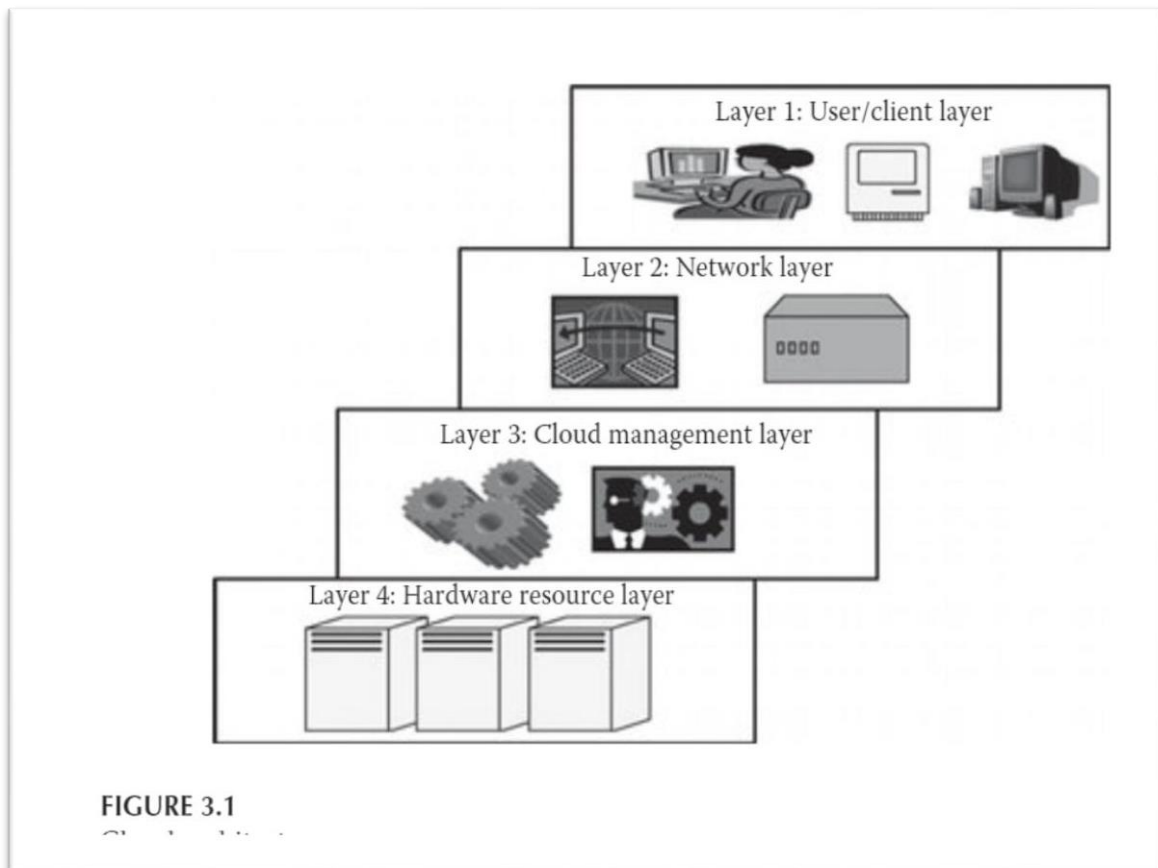
Management is used to manage components such as application, service, runtime cloud, storage, infrastructure, and other security issues in the backend and establish coordination between them.

8. Security:

Security is an in-built back end component of cloud computing. It implements a security mechanism in the back end.

9. Internet :

The Internet is medium through which front end and back end can interact and communicate with each other.

Layers of cloud computing architectures:

There are four layers in cloud computing architectures:

1. User / Client Layer
2. Network Layer
3. Cloud Management Layer
4. Hardware Resource Layer

Layer 1 (User/ Client Layer) :

- It is the topmost layer of cloud architecture.
- The actors of this layer are the end users and the clients.
- The clients are the devices that can either be a **thick client**, a **thin client**, or a mobile device capable of accessing the web application.
- Here a **thick client** is a computer that is capable of working independently. It is the one with adequate processing capability.
- Whereas, the **thin client** is the one with the low processing capability. It depends on another device for complete functionality.
- At layer 1, the user and client initiate to connect themselves to the cloud.
- Accessing cloud applications is similar to accessing web applications.
- Access to both types of applications requires the internet or other computer networks.
- Although the internal properties of the cloud application are different from web applications. Hence, we require layer 1 to set up the connection between the user/client and the cloud.

Layer 2 (Network Layer) :

- Layer 2 or the network layer is responsible for connecting users to the cloud.
- The absence of a network layer wouldn't work for the cloud.
- Because the entire infrastructure of the cloud depends on the connection where it offers the services to the users. For accessing the public cloud we make connections using the internet.
- For accessing the private cloud, the connectivity is established using the local area network, i.e. (LAN). Each kind of connection requires a minimum bandwidth specified by the cloud providers.
- However, the network layer is beyond the cloud SLA i.e. service level agreement. SLA promises a minimum level of services to cloud customers.
- SLA does not consider the connection between the cloud and the user for quality of service (QoS).

Layer 3 (Cloud Management Layer) :

- The layer is responsible for managing the cloud services. It includes the software that manages the working of the cloud. The software used at this layer can be: Cloud OS (Operating System).
- Software is responsible for creating an interface between the cloud and its users.
- Software for managing the resources.
- The software at the cloud management layer is responsible for: Managing resources (scheduling the cloud services, provisioning resources, etc.).

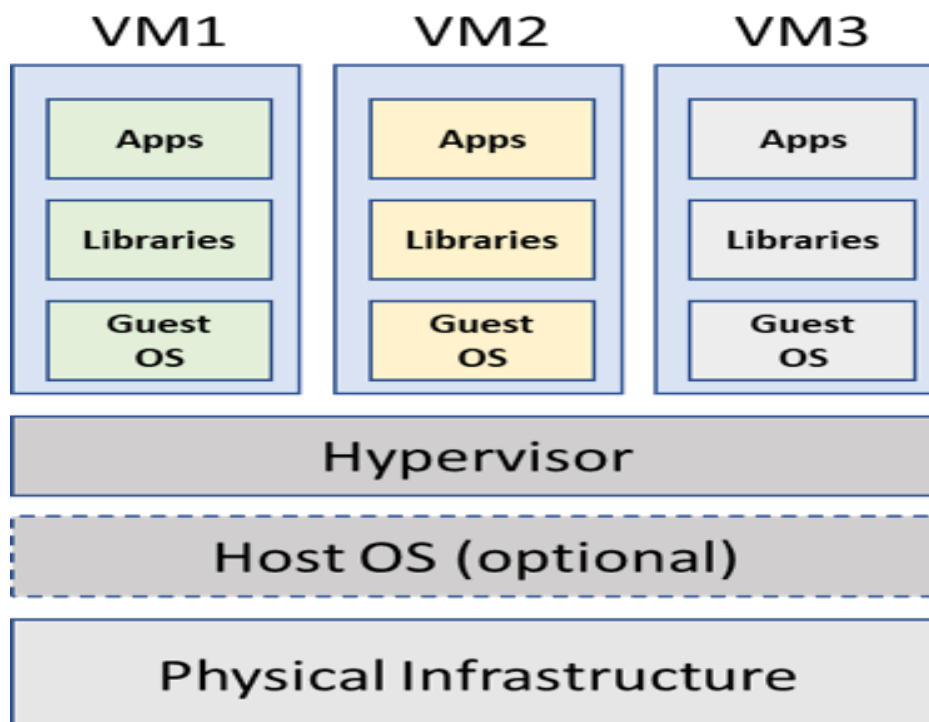
- Optimization (Server consolidation i.e. reducing the total number of servers of the company by making efficient use of server resources, workload consolidation i.e. integrating multiple computational operations onto fewer platforms, storage consolidation i.e. centralizing the data storage and minimizing the time required to access data) Internal cloud governance, the network layer comes under close surveillance of SLA.
- Thus, the operations that take place at this layer will affect the agreement that SLA decide between the cloud users and the providers. So, a delay in connecting to the cloud would also result in a violation of SLA. In such cases, the providers have to pay penalties. The SLA purviews are both public and private and could for any violation.

Layer 4 (Hardware Resource Layer) :

- It defines the provision of hardware resources.
- Layer 4 of a public cloud is the huge data centre.
- Whereas layer 4 of the private cloud is constructed with a huge collection of hardware resources.
- These resources are interconnected to each other and are present at a specific location or it may be a high configuration system. The hardware resource layer comes under the purview of service layer agreements(SLAs). We can even say that the layer governs SLAs.
- As we have discussed previously that the user must be able to access the cloud services as quick as possible.
- If there occurs any discrepancy(Something is wrong) in the provisioning of the hardware resources then the provider might have to pay the penalty.
- So, the hardware resource layer of any cloud architecture must consist of data centres with high-speed network connections.
- The data centre must have a highly efficient algorithm that facilitates data transfer from the data centre to the manager.
- A single cloud can have multiple data centres and even multiple clouds can share a single data centre. So, this is the layered descript of a cloud architecture.
- Every cloud architecture must strictly flow this layering.

Virtualization –Definition:

- **Virtualization** is the "creation of a virtual (rather than actual) version of something, such as a server, a desktop, a storage device, an operating system or network resources".
- In other words, Virtualization is a technique, which allows sharing a single physical instance of a resource or an application among multiple customers and organizations.
- **Virtualization** can be defined as a process that enables the creation of a virtual version of a *desktop, operating system, network resources, or server*.
- The machine on which the virtual machine is going to create is known as **Host Machine**
- Virtual machine is referred as a **Guest Machine**.



- It helps reduce the **space** or **cost** involved with the resource.
- This technique enables the **end-user** to run multiple desktop operating systems and applications simultaneously on the same hardware and software.

Hypervisors

Hypervisors are software tools used to create the VMs, and they produce the virtualization of various hardware resources such as CPU, storage, and networking devices. They are also called virtual machine monitor (VMM) or virtualization managers.

They help in the virtualization of cloud data centers (DCs). The various hypervisors used are **VMware, Xen, Hyper-V, KVM**, etc. Hypervisors help to run multiple OSs concurrently on a physical system sharing its hardware.

Hypervisors are of mainly two types:

1. Type 1 hypervisor: (bare metal hypervisors)

- This type of hypervisor runs or install directly on the host computer's hardware in order to control the hardware resources and also to manage the guest OSs.
- This is also known as native or bare-metal hypervisors.
- Examples include **VMware ESXi, Citrix XenServer, and Microsoft Hyper-V hypervisor**.

2. Type 2 hypervisor (hosted hypervisors)

- This type of hypervisor runs or install within a formal/Host OS environment.
- In this type, the hypervisor runs as a distinct second layer while the guest OS runs as a third layer above the hardware.
- This is also known as the hosted hypervisors.
- Examples include **VMware Workstation and VirtualBox**.

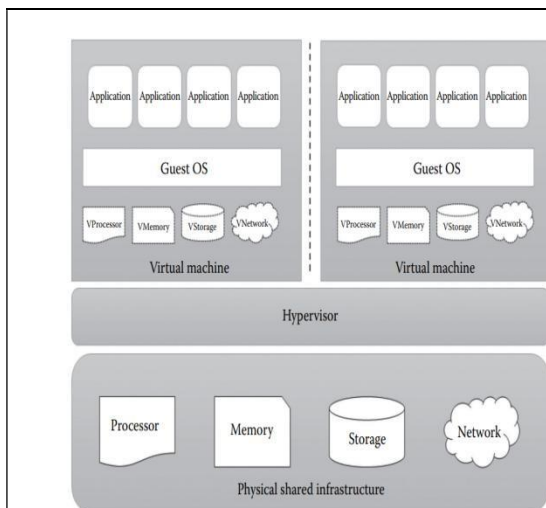


FIGURE 7.13
Type 1 or bare metal hypervisor.

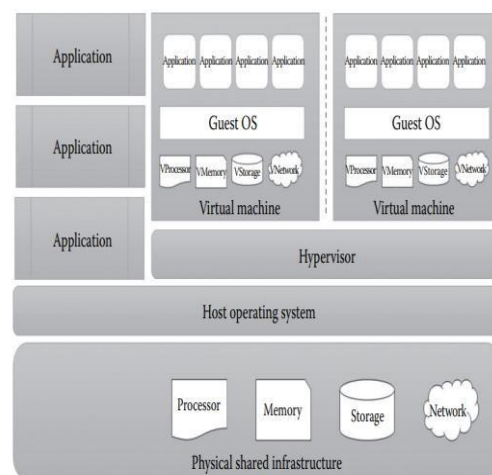


FIGURE 7.14
Type 2 or hosted hypervisor.

Features of Virtualization (in Cloud computing)

1. Resource Isolation (Efficient Resource use):

- **Feature:** Virtualization enables resource isolation, meaning each virtual instance runs independently without interfering with others.
- **Example:** In a cloud data center, multiple virtual servers can coexist on the same physical hardware. They are isolated from each other, ensuring one virtual server's actions do not affect the others.

2. Scalability:

- **Feature:** Virtualization allows for easy scaling of resources up or down to meet changing demands.
- **Example:** A website hosted on a virtual server can quickly handle increased traffic by adding more virtual instances or resources as needed, and then scale down during periods of low demand.

3. Server Consolidation:

- **Feature:** Multiple virtual servers can be consolidated onto a single physical server, optimizing resource utilization.
- **Example:** A company can run multiple virtual servers on a single physical server, reducing the number of physical machines needed, leading to cost savings in terms of hardware and energy.

4. Resource Pooling:

- **Feature:** Resources such as CPU, memory, and storage can be pooled and allocated dynamically as needed.
- **Example:** In a cloud environment, multiple users share the same physical resources. The cloud provider manages and allocates these resources based on demand.

5. Cost-Efficiency:

- **Feature:** Virtualization reduces the need for extensive hardware investments, leading to cost savings.
- **Example:** Instead of buying and maintaining multiple physical servers, a business can rent virtual resources from a cloud provider, paying only for what is used.

6. Live Migration:

- **Feature:** Virtual machines can be moved from one physical server to another with minimal downtime.
- **Example:** Cloud providers use **live migration** to balance server loads, perform hardware maintenance, or optimize resource usage without affecting the running VMs.

7. Disaster Recovery:

- **Feature:** Virtualization simplifies disaster recovery by allowing for easy backup and restoration of virtual instances.
- **Example:** In case of a hardware failure, a backup of a virtual machine can be quickly restored on another server, minimizing downtime.

8. Operating System Independence:

- **Feature:** Virtualization allows you to run different operating systems on the same physical hardware.
- **Example:** You can host a Windows-based virtual machine alongside a Linux-based one on the same physical server, providing flexibility for diverse application requirements.

9. Resource Monitoring and Management:

- **Feature:** Virtualization provides tools for monitoring and managing resources to optimize performance and cost.
- **Example:** Cloud providers offer dashboards and management interfaces to track resource usage, identify bottlenecks, and make real-time adjustments.

10. Reduced Energy Consumption:

- **Feature:** Virtualization contributes to energy efficiency by consolidating workloads onto fewer physical servers.
 - **Example:** By running multiple virtual servers on a single physical machine, data-centers can reduce energy consumption and lower their carbon footprint.
-

Types of Virtualizations-

1. Hardware Virtualization
2. Server Virtualization
3. Application Virtualization
4. Storage Virtualization
5. Operating System Virtualization

Hardware Virtualization:

- When the virtual machine software or virtual machine manager (VMM) is directly **installed** on the **hardware system** is known as hardware virtualization.
- The main job of hypervisor is to control and monitoring the processor, memory and other hardware resources. After virtualization of hardware system we can

install different operating system on it and run different applications on those OS.

Usage:

Hardware virtualization is mainly done for the server platforms, because controlling virtual machines is much easier than controlling a physical server.

- **Definition:** *Hardware virtualization* allows you to create multiple virtual machines (VMs) on a single physical server, each with its own virtualized hardware resources.
- **Real-time Example:** **VMware vSphere/ESXi** is a popular hypervisor that provides hardware virtualization. You can install **ESXi** on a physical server and create multiple virtual machines on it. Each VM operates independently with its own virtual **CPU, RAM, and storage**.

Server Virtualization:

- **When the virtual machine software or virtual machine manager (VMM) is directly installed on the Server system is known as server virtualization.**

Usage:

Server virtualization is done because a single physical server can be divided into multiple servers on the demand basis and for balancing the load.

Server Virtualization:

- **Definition:** **Server virtualization is a subset of hardware virtualization focused on creating multiple virtual server instances on a single physical server.**
- **Real-time Example:** Consider a company that needs multiple web servers for different projects. With server virtualization using **VMware** or **Microsoft Hyper-V**, they can run multiple virtual servers on a single physical machine. Each virtual server can host a different website or application, optimizing resource utilization.

Application Virtualization:

- This can be defined as the type of Virtualization that enables the end-user of an application to have remote access.
- This is achieved through a server.
- This server has all personal information and other applicable characteristics required to use the application.
- The server is accessible through the internet, and it runs on a local workstation.
- With Application virtualization, an end-user can run two different versions of the same software or the same application.
- Application virtualization is offered through packaged software or a hosted application.

Application Virtualization:

- **Definition:** Application virtualization separates applications from the underlying operating system, allowing them to run in isolated environments.
- **Real-time Example:** *Citrix XenApp* is an application virtualization solution. It allows businesses to host and deliver applications to users without installing them on local devices. This simplifies software management and ensures compatibility. For instance, a company can host **Microsoft Office applications** centrally and deliver them to remote users.

Storage Virtualization:

- Storage virtualization is the process of grouping the physical storage from multiple network storage devices so that it looks like a single storage device.
- Storage virtualization is also implemented by using software applications.
- In **storage virtualization**, multiple physical hard drives are combined into a single virtualized storage environment.
- To various users, this is simply called cloud storage, and it could be a private storage, such that it is hosted by a company, or a public storage, such that it is hosted outside of a company like **DropBox**, or a mixed approach of the two.

Usage:

- **Storage virtualization is mainly done for back-up and recovery purposes.**

Storage Virtualization:

- **Definition:** Storage virtualization abstracts and pools storage resources from different physical devices into a single, unified storage pool.
- **Real-time Example: *Storage Area Networks* (SANs)** often use storage virtualization. A SAN can combine storage devices from various vendors into a single storage pool, making it easier to manage, scale, and allocate storage resources.

Operating System Virtualization:

- When the virtual machine software or virtual machine manager (VMM) is installed on the Host operating system instead of directly on the hardware system is known as operating system virtualization.

Usage:

Operating System Virtualization is mainly used for testing the applications on different platforms of OS.

Operating System Virtualization (Containerization):

- **Definition:** Operating system virtualization, or containerization, isolates(separate) applications in lightweight, portable containers that share the host OS kernel.
 - **Real-time Example: Docker** is a widely used containerization platform. Developers can create containers containing their applications and their dependencies, and these containers can run consistently across different environments. For instance, a developer can create a container for a web application, and it will run the same way on a developer's laptop and in a production server.
-

Virtualization and Cloud Computing:

Many users of current IT solutions consider the technologies **virtualization** and **cloud computing** as the same.

But both technologies are actually different, or in other words, we can say virtualization is not cloud computing.

We can prove this claim with the following parameters:

1. **Type of service:** Generally, virtualization offers more infrastructure services rather than **platform** and **application services**. But cloud computing offers all *infrastructure (IaaS)*, **platform (PaaS)**, and **software (SaaS)** services.
 2. **Service delivery:** The service delivery in cloud computing is ondemand and allows the end users to use the cloud services as per the need. But virtualization is not made for on-demand services.
 3. **Service provisioning:** In cloud computing, automated and self-service provisioning is possible for the end users, whereas in virtualization, it is not possible and a lot of manual work is required from the providers or system administrator to provide services to the end users.
 4. **Elasticity:** One of the important characteristics that differentiate cloud computing from virtualization is elasticity. In cloud computing, we can add or remove the infrastructure dynamically according to the need, and adding or removing the infrastructure is automatic. But virtualization fails to provide elasticity as stopping and starting a VM is manual and is also difficult.
-

Pros and Cons of Virtualization

Pros of Virtualization in Cloud Computing:

1. Resource Efficiency:

- **Pro:** Virtualization allows better utilization of physical resources. Multiple virtual machines (VMs) can run on a single physical server.
- **Example:** In a cloud data center, many VMs with varying workloads can share the same server hardware, reducing resource wastage.

2. Scalability:

- **Pro:** Cloud providers can quickly provision and de-provision VMs based on demand, making it easy to scale resources up or down.
- **Example:** An e-commerce website can scale up during holiday sales and scale down during slower periods without investing in new physical servers.

3. Isolation and Security:

- **Pro:** VMs are isolated from one another, enhancing security. A security breach in one VM doesn't necessarily affect others.
- **Example:** In a cloud environment, sensitive financial data can be stored in one VM, while a public-facing website is in another VM.

4. Disaster Recovery:

- **Pro:** VM snapshots and backups make disaster recovery easier. VMs can be quickly restored to a previous state.
- **Example:** In case of data corruption, a cloud-based application can be restored to a known-good state within minutes.

5. Cost Savings:

- **Pro:** Virtualization reduces hardware costs and power consumption, leading to cost savings.
- **Example:** Businesses can avoid investing in and maintaining on-premises servers, opting for more cost-effective cloud-based infrastructure.

6. Flexibility:

- **Pro:** Virtualization allows for flexibility in managing resources. You can allocate, reallocate, or resize VMs as needed.

- **Example:** During a software development project, developers can request VMs with specific configurations, optimizing their development and testing environments.

7. Portability:

- **Pro:** Virtualized workloads are often portable, which means you can move them between different cloud providers or on-premises environments.
- **Example:** If a company decides to change cloud providers, they can migrate their VMs, making it easier to transition.

Cons of Virtualization in Cloud Computing:

1. Overhead:

- **Con:** Virtualization adds some overhead due to the hypervisor, which can slightly reduce the performance of VMs.
- **Example:** In a virtualized environment, a high-performance application might experience a minor decrease in speed due to the virtualization layer.

2. Complexity:

- **Con:** Managing virtualized resources can be complex, especially when dealing with a large number of VMs.
- **Example:** Cloud administrators may need to use management tools to orchestrate and maintain numerous VMs, which can be challenging.

3. Security Risks:

- **Con:** While VM isolation is a pro, there can be security risks if VMs are not properly configured and secured.
- **Example:** Misconfigured VMs may expose sensitive data, leading to potential breaches.

4. Licensing Costs:

- **Con:** Virtualization software often comes with licensing costs, which can add to the overall cloud expenses.
- **Example:** A company using VMware virtualization in the cloud would need to pay for VMware licenses in addition to cloud service fees.

5. Vendor Lock-In:

- **Con:** Adopting a specific virtualization technology may lead to vendor lock-in, making it difficult to migrate to a different cloud provider.
 - **Example:** If a company heavily relies on a cloud provider's proprietary virtualization technology, it may face challenges if it decides to switch to a different cloud provider.
-

Technology Examples:

Xen: Paravirtualization,
VMware: Full Virtualization,
Microsoft Hyper-V.

Technology examples:

There is a wide range of virtualization technologies available especially for virtualizing computing environments. In this section, we discuss the most relevant technologies and approaches utilized in the field.

Xen: Paravirtualization:

- Xen is an open source initiative implementing a virtualization platform based on paravirtualization.
- Initially developed by a group of researchers at the University of Cambridge.

Xen:

Overview: Xen is an open-source hypervisor, which is a type of software that enables multiple operating systems to run on a single physical machine simultaneously.

Role of Xen: Xen acts as a mediator between the computer's hardware and the various operating systems running on it. It allocates resources, manages communication, and ensures each operating system operates independently.

What is Xen?

Xen is a type-1 hypervisor, which means it's a software platform that allows *multiple operating systems* to run on the same hardware simultaneously.

In Simple Terms: Imagine Xen as a traffic cop for your computer. It helps different operating systems (like Windows and Linux) to share the same computer without getting in each other's way.

Xen : Paravirtualization

- **Definition:** Paravirtualization is a virtualization technique where the guest operating system is modified to be aware of the virtualized environment.
- It's a technique where the operating system is modified to work more efficiently with the hypervisor (like **Xen**).
- Xen follows **paravirtualization** technology. Paravirtualization is a virtualization technique that involves modifying the operating system (OS) running inside a virtual machine (VM) to make it aware of its virtualized environment. Unlike full virtualization, where the guest OS is unaware of the virtualization layer and runs unmodified, paravirtualization requires changes to the guest OS to improve **performance, security, and efficiency** and also reduce **overhead**.
- Example: **XenoLinux** is a modified Linux OS that runs on Xen hypervisor.

Advantages of Paravirtualization:

- **Performance:** Paravirtualized guests can achieve better performance compared to full virtualization because they are aware of the virtualization layer.
- **Efficiency:** It enhances communication between the operating system and the hypervisor, making it more efficient.
- **Resource Utilization:** Paravirtualization allows for optimized resource utilization.

How Xen Uses Paravirtualization:

- **Modified Guest OS:** In a paravirtualized setup with Xen, the guest operating systems are adjusted or modified to be Xen-aware.
- **Communication Optimization:** The communication between the guest OS and Xen is streamlined, bypassing certain traditional virtualization layers for faster and more direct interaction.
- **Hypercalls:** Instead of using traditional hardware calls (syscalls), the guest OS in a paravirtualized environment makes hypercalls, which are calls specifically designed for communication with the hypervisor.

Benefits of Xen paravirtualization**1. Faster and Smoother Performance:**

- **Explanation:** Xen paravirtualization reduces the time it takes for different parts of your computer to communicate with each other, making everything run faster.
- **Example:** Think of it like a group of friends talking with a walkie-talkie (paravirtualization) instead of shouting across a large field. The messages get through quicker, and everyone can respond faster.

2. Efficient Use of Resources:

- **Explanation:** Xen helps your computer use its memory, processor, and other resources more efficiently, avoiding wastage.
- **Example:** Imagine you have a pizza and you want to share it with friends. Instead of giving each friend a whole pizza (which might be too much), Xen helps you cut the pizza into just the right number of slices for each friend, avoiding any leftover slices.

3. Strong Isolation for Security:

- **Explanation:** Xen keeps different parts of your computer separated so that if one part has a problem, it doesn't affect the others. This helps keep your computer secure.
- **Example:** It's like having different rooms in your house for different activities. If something spills in the kitchen, it doesn't affect what you're doing in the living room.

4. Compatibility with Various Operating Systems:

- **Explanation:** Xen works well with many different types of operating systems, allowing you to run various programs without any issues.
- **Example:** Think of it like having a magical translator that helps you understand and communicate with friends who speak different languages. Xen is like that translator, helping different parts of your computer (operating systems) understand each other.

VMware: Full Virtualization**1. VMware:**

- **Overview:** VMware is a company that provides virtualization software, including hypervisors that enable full virtualization.
- **Hypervisor:** VMware uses a type-1 hypervisor, which is a layer of software that runs directly on the hardware and allows multiple operating systems to run on the same physical machine.

2. Full Virtualization:

- **Definition:** Full virtualization is a virtualization technique where the guest operating systems are not aware that they are running in a virtualized environment. They operate as if they are running on dedicated hardware.
- **Advantages:** Full virtualization provides a high level of isolation between virtual machines (VMs), allowing different operating systems to run concurrently without modification.

3. How VMware Achieves Full Virtualization:

- **Virtual Machine Monitor (VMM):** VMware's hypervisor acts as a Virtual Machine Monitor (VMM) that creates and manages virtual machines.
- **Hardware Emulation:** In full virtualization, the hypervisor emulates the underlying hardware, presenting a virtual set of hardware components to each VM. This allows unmodified operating systems to run within the virtual machines.

4. Real-World Example:**Virtualized Data Center:**

- **Traditional Data Center:** Imagine a traditional data center with multiple physical servers, each running its own operating system.
- **VMware Full Virtualization:** Now, with VMware, you can consolidate these physical servers into a single, more efficient server by running multiple virtual machines on it. Each VM operates independently, running its own operating system, but they all share the same underlying hardware.
- **Scenario:** If you have one virtual machine running Windows and another running Linux on the same physical server using VMware:

- **Windows VM:** It behaves just like a standalone Windows server, unaware that it's sharing resources with other VMs.
- **Linux VM:** Similarly, the Linux VM operates as if it's running on dedicated hardware.

- **Benefits of VMware Full Virtualization:**

- **Resource Optimization:** Full virtualization allows efficient use of hardware resources by running multiple virtual machines on a single physical server.
- **Isolation:** Each virtual machine is isolated from others, providing a secure environment for different operating systems and applications.
- **Flexibility and Mobility:** VMs can be easily moved or copied between different physical servers, providing flexibility in resource allocation and workload management.

- **Use Case Example:**

- **Cloud Computing Infrastructure:**

- **Scenario:** Imagine a cloud service provider using VMware for full virtualization in their data centers.
- **Benefits:**
 - They can offer virtual machines with different operating systems (Windows, Linux) to their customers.
 - Customers can run their applications in isolated VMs without worrying about interference from other users.
 - The cloud provider can efficiently allocate resources, scale services, and manage workloads across their infrastructure.

VMware's full virtualization allows multiple operating systems to run on a single physical machine without modification to the guest OS. This is achieved through hardware emulation, binary translation, and the use of a hypervisor, providing a flexible and efficient way to manage computing resources in data centers.

Microsoft Hyper-V

1. Overview of Microsoft Hyper-V:

- **What is Hyper-V?**

Hyper-V is a virtualization platform developed by Microsoft, allowing users to create and manage virtual machines on Windows-based systems.

Hypervisor Type: Hyper-V is a *type-1 hypervisor*, meaning it runs directly on the hardware to control and manage virtualized resources.

2. Key Features:

1. Virtual Machine Management:

- **Explanation:** Hyper-V allows you to create, configure, and manage virtual machines (VMs) on a single physical server.
- **Example:** Imagine having a powerful computer, and within it, you can create multiple computers, each running its own operating system. This helps in running different applications or services on isolated virtual environments.

2. Snapshot and Checkpoints:

- **Explanation:** Hyper-V enables you to take snapshots or checkpoints of virtual machines at specific points in time, allowing you to revert to a previous state if needed.
- **Example:** It's like taking a snapshot of your computer before installing new software. If something goes wrong, you can go back to the snapshot where everything was working fine.

3. Live Migration:

- **Explanation:** Hyper-V supports live migration, allowing you to move running virtual machines between different physical servers without any downtime.
- **Example:** Imagine you are playing a game on your computer, and without pausing it, you can seamlessly switch to another computer without losing progress. HyperV allows a similar smooth transition for running applications.

4. Integration Services:

- **Explanation:** Integration Services enhance communication between the host system and virtual machines, improving performance and providing additional functionality.
- **Example:** It's like having a language translator (Integration Services) between you (host system) and your friend (virtual machine), ensuring clear communication and efficient collaboration.

5. Dynamic Memory Allocation:

- o **Explanation:** Hyper-V allows the dynamic allocation of memory to virtual machines based on demand, optimizing resource utilization.
- o **Example:** Think of it as having a backpack that automatically adjusts its size depending on what you put inside. Hyper-V dynamically adjusts the memory allocated to virtual machines based on their needs.

Real-Time Examples:

Corporate IT Infrastructure:

- **Scenario:** A company uses Hyper-V to virtualize its servers, hosting various applications and services.
- **Example:** The company runs its email server, file-sharing system, and customer relationship management (CRM) software on separate virtual machines, allowing for easy management, scalability, and efficient resource utilization.

Disaster Recovery:

- **Scenario:** An organization uses Hyper-V to create replica virtual machines in a different geographical location for disaster recovery.
- **Example:** If the primary data center faces an issue, the organization can quickly switch to the replicated virtual machines in another location using Hyper-V's capabilities for disaster recovery.

Development and Testing:

- **Scenario:** Developers use Hyper-V to create virtual environments for testing new software and applications.
- **Example:** A developer can have multiple virtual machines with different operating systems to test software compatibility. If something goes wrong, they can easily revert to a previous state using Hyper-V checkpoints.

Cloud Hosting:

- **Scenario:** Cloud service providers utilize Hyper-V to host virtual machines for their customers.
- **Example:** When you rent a virtual server from a cloud provider, it's likely running on a Hyper-V infrastructure. The provider can allocate and manage virtual resources efficiently, offering flexibility and scalability to their customers.